Class 10 Mathematics — Exercise 2.1

Basic Concept

Quadratic Equation

An equation whose **highest power (degree) is 2** is called a quadratic equation.

- $2x^2 + 5x + 3 = 0$ ✓ (highest power 2)
- $2x^2 + 5x + 3$ × (not an equation)
- $2x^3 + 5x + 3 = 0$ × (highest power 3)

Parts of an equation: **Variable** (unknown), **Power/Degree**, **Constant**.

Standard Form of Quadratic Equation

$$\boxed{ax^2 + bx + c = 0, \qquad a \neq 0}$$

- $2x^2 + 5x + 3 = 0$ ✓ (standard form)
- $2x^2 + 5x = -3$ × (not in standard form)

Methods to Solve Quadratic Equation

1. Factoring Method → Q2
2. Completing Square Method → Q3
3. Using Quadratic Formula → Q4

Question 1

Write the following quadratic equation in standard form:

(i) $(x + 2)(x - 3) = 5$

$$x(x - 3) + 2(x - 3) - 5 = 0$$

$$x^2 - 3x + 2x - 6 - 5 = 0$$

$$\boxed{x^2 - x - 11 = 0}$$

(ii) $(x - 5)^2 - (2x + 4)^2 = 7$

$$[(x)^2 + (5)^2 - 2(x)(5)] - [(2x)^2 + (4)^2 + 2(2x)(4)] - 7 = 0$$

$$(x^2 + 25 - 10x) - (4x^2 + 16 + 16x) - 7 = 0$$

$$x^2 + 25 - 10x - 4x^2 - 16 - 16x - 7 = 0$$

$$-3x^2 - 26x + 2 = 0$$

$$-(3x^2 + 26x - 2) = 0$$

$$\boxed{3x^2 + 26x - 2 = 0} \qquad \left(\because \frac{0}{-1} = 0\right)$$

(iii) $x = x(x-1)$

$$x = x^2 - x$$

$$0 = x^2 - x - x$$

$$0 = x^2 - 2x$$

$$\boxed{x^2 - 2x = 0}$$

Question 2

Solve the following equations by factoring method.

Factoring Method (Factorization):

1. Write the equation in standard form, then make factors.
2. Use **zero product property**: if $ab = 0$ then $a = 0$ or $b = 0$.

(i) $(x-1)(x-4) = 0$

$$x - 1 = 0 \quad , \quad x - 4 = 0$$

$$x = 1 \quad , \quad x = 4$$

$$\boxed{\text{S.S} = \{1, 4\}}$$

(ii) $x^2 - 2x + 1 = 0$

Split the middle term: $1 = (-1) \times (-1)$ and $(-1) + (-1) = -2$

$$x^2 - x - x + 1 = 0$$

$$x(x-1) - 1(x-1) = 0$$

$$(x-1)(x-1) = 0$$

$$x - 1 = 0 \quad \Big| \quad x - 1 = 0$$

$$x = 1 \quad \Big| \quad x = 1$$

$$\boxed{\text{S.S} = \{1\}}$$

(iii) $x^2 - 7x - 8 = 0$

Split the middle term: $-8 = 8 \times 1 \times, 4 \times 2 \times \Rightarrow (-8) \times 1$ ✓ and $-8 + 1 = -7$

$$x^2 - 8x + x - 8 = 0$$

$$x(x - 8) + 1(x - 8) = 0$$

$$(x + 1)(x - 8) = 0$$

$$x + 1 = 0 \quad \Big| \quad x - 8 = 0$$

$$x = -1 \quad \Big| \quad x = 8$$

$$\boxed{\text{S.S} = \{-1, 8\}}$$

(iv) $x^2 - 4x + 4 = (2x - 7)^2$

Using $(a - b)^2 = a^2 + b^2 - 2ab$:

$$x^2 - 4x + 4 = (2x)^2 + (7)^2 - 2(2x)(7)$$

$$x^2 - 4x + 4 = 4x^2 + 49 - 28x$$

$$0 = 4x^2 - x^2 - 28x + 4x + 49 - 4$$

$$0 = 3x^2 - 24x + 45$$

$$3x^2 - 24x + 45 = 0$$

$$3(x^2 - 8x + 15) = 0$$

$$x^2 - 8x + 15 = 0 \qquad \left(\because \frac{0}{3} = 0\right)$$

Split the middle term: $15 = 15 \times 1 \times, (-5) \times (-3)$ ✓ and $-5 - 3 = -8$

$$x^2 - 5x - 3x + 15 = 0$$

$$x(x - 5) - 3(x - 5) = 0$$

$$(x - 3)(x - 5) = 0$$

$$x-3=0 \quad \Big| \quad x-5=0$$

$$x=3 \quad \Big| \quad x=5$$

$$\boxed{\text{S.S} = \{3, 5\}}$$

(v) $\left(2x+\frac{7}{4}\right)^2 = \frac{48x^2+529}{16}$

$$\left(\frac{8x+7}{4}\right)^2 = \frac{48x^2+529}{16}$$

$$\frac{(8x)^2+(7)^2+2(8x)(7)}{16} = \frac{48x^2+529}{16}$$

$$64x^2+49+112x = 48x^2+529$$

$$64x^2-48x^2+112x+49-529=0$$

$$16x^2+112x-480=0$$

$$16(x^2+7x-30)=0$$

$$x^2+7x-30=0$$

Split the middle term: $-30 = 30\times 1\ \times,\ 10\times(-3)$ ✓ and $10-3=7$

$$x^2-3x+10x-30=0$$

$$x(x-3)+10(x-3)=0$$

$$(x+10)(x-3)=0$$

$$x+10=0 \quad \Big| \quad x-3=0$$

$$x=-10 \quad \Big| \quad x=3$$

$$\boxed{\text{S.S} = \{-10, 3\}}$$

Question 3

Solve the following equations by completing the square method.

Steps of Completing Square Method:

1. Write in standard form; coefficient of x^2 must be **1**.
2. Shift the constant to R.H.S.
3. Add the square of half of the coefficient of x on both sides, i.e. $\left(\frac{b}{2}\right)^2$.
4. Take square root on both sides.
5. Find the values of x.

(i) $x^2 + 4x - 32 = 0$

$$x^2 + 4x = 32 \qquad \left(\frac{4}{2} = 2\right)$$

$$x^2 + 4x + (2)^2 = 32 + (2)^2$$

$$(x+2)^2 = 32 + 4$$

$$(x+2)^2 = 36$$

$$\sqrt{(x+2)^2} = \sqrt{36}$$

$$x + 2 = \pm 6$$

$$x + 2 = 6 \quad \Big| \quad x + 2 = -6$$

$$x = 6 - 2 \quad \Big| \quad x = -6 - 2$$

$$x = 4 \quad \Big| \quad x = -8$$

$$\boxed{\text{S.S} = \{4, -8\}}$$

(ii) $x^2 + 8x = 0$

$$\left(\frac{8}{2} = 4\right)$$

$$x^2 + 8x + (4)^2 = 0 + (4)^2$$

$$(x+4)^2 = 0 + 16$$

$$(x+4)^2 = 16$$

$$\sqrt{(x+4)^2} = \sqrt{16}$$

$$x + 4 = \pm 4$$

$$x + 4 = 4 \quad \Big| \quad x + 4 = -4$$

$$x = 4 - 4 \quad \Big| \quad x = -4 - 4$$

$$x = 0 \quad \Big| \quad x = -8$$

$$\boxed{\text{S.S} = \{0, -8\}}$$

(iii) $x^2 + 6x - 9 = 0$

$$x^2 + 6x = 9 \qquad \left(\frac{6}{2} = 3\right)$$

$$x^2 + 6x + (3)^2 = 9 + (3)^2$$

$$(x+3)^2 = 9 + 9$$

$$(x+3)^2 = 18$$

$$\sqrt{(x+3)^2} = \sqrt{18} \qquad \left(\because \sqrt{18} = \sqrt{9 \times 2} = 3\sqrt{2}\right)$$

$$x + 3 = \pm 3\sqrt{2}$$

$$x = -3 \pm 3\sqrt{2}$$

$$\boxed{\text{S.S} = \{-3 \pm 3\sqrt{2}\}}$$

(iv) $3x^2 + 12x + 8 = 0$

Divide each term by 3 (to make coefficient of x^2 equal to 1):

$$\frac{3x^2}{3} + \frac{12x}{3} + \frac{8}{3} = \frac{0}{3}$$

$$x^2 + 4x + \frac{8}{3} = 0$$

$$x^2 + 4x = -\frac{8}{3} \qquad \left(\frac{4}{2} = 2\right)$$

$$x^2 + 4x + (2)^2 = -\frac{8}{3} + (2)^2$$

$$(x+2)^2 = -\frac{8}{3} + 4 = \frac{-8+12}{3}$$

$$(x+2)^2 = \frac{4}{3}$$

$$\sqrt{(x+2)^2} = \sqrt{\frac{4}{3}}$$

$$x + 2 = \pm\frac{2}{\sqrt{3}}$$

$$x = -2 \pm \frac{2}{\sqrt{3}} = \frac{-2\sqrt{3} \pm 2}{\sqrt{3}}$$

$$\boxed{\text{S.S} = \left\{ \frac{-2\sqrt{3} \pm 2}{\sqrt{3}} \right\}}$$

(v) $x^2 + x + 1 = 0$

$$x^2 + x = -1 \qquad \left(\frac{1}{2}\right)$$

$$x^2 + x + \left(\frac{1}{2}\right)^2 = -1 + \left(\frac{1}{2}\right)^2$$

$$\left(x + \frac{1}{2}\right)^2 = -1 + \frac{1}{4} = \frac{-4+1}{4}$$

$$\left(x + \frac{1}{2}\right)^2 = -\frac{3}{4}$$

$$\sqrt{\left(x + \frac{1}{2}\right)^2} = \sqrt{-\frac{3}{4}}$$

$$x + \frac{1}{2} = \pm\frac{\sqrt{-3}}{2}$$

$$x = -\frac{1}{2} \pm \frac{\sqrt{-3}}{2}$$

$$x = \frac{-1 \pm \sqrt{-3}}{2}$$

$$\boxed{\text{S.S} = \left\{ \frac{-1 \pm \sqrt{-3}}{2} \right\}}$$

(vi) $4x^2 - 8x - 5 = 0$

Divide each term by 4:

$$\frac{4x^2}{4} - \frac{8x}{4} - \frac{5}{4} = \frac{0}{4}$$

$$x^2 - 2x - \frac{5}{4} = 0$$

$$x^2 - 2x = \frac{5}{4} \qquad \left(\frac{2}{2} = 1\right)$$

$$x^2 - 2x + (1)^2 = \frac{5}{4} + (1)^2$$

$$(x-1)^2 = \frac{5}{4} + \frac{1}{1} = \frac{5+4}{4}$$

$$(x-1)^2 = \frac{9}{4}$$

$$\sqrt{(x-1)^2} = \sqrt{\frac{9}{4}}$$

$$x - 1 = \pm\frac{3}{2}$$

$$x = \frac{1}{1} \pm \frac{3}{2} = \frac{2 \pm 3}{2}$$

$$x = \frac{2+3}{2} = \frac{5}{2} \quad \Big| \quad x = \frac{2-3}{2} = -\frac{1}{2}$$

$$\boxed{\text{S.S} = \left\{\frac{5}{2}, -\frac{1}{2}\right\}}$$

Question 4

Solve the following equations by quadratic formula:

$$\boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

Compare the equation with $ax^2 + bx + c = 0$ to get a, b, c. For example, in $2x^2 - 5x + 4 = 0$: $a = 2,\ b = -5,\ c = 4$.

Derivation of Quadratic Formula

$$ax^2 + bx + c = 0$$

Divide by a:

$$\frac{ax^2}{a} + \frac{bx}{a} + \frac{c}{a} = \frac{0}{a}$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

$$x^2 + \frac{b}{a}x = -\frac{c}{a} \qquad \left(\because \frac{b}{a} \times \frac{1}{2} = \frac{b}{2a}\right)$$

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2$$

$$\left(x + \frac{b}{2a}\right)^2 = -\frac{c}{a} + \frac{b^2}{4a^2} = \frac{b^2}{4a^2} - \frac{c}{a}$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

$$\sqrt{\left(x + \frac{b}{2a}\right)^2} = \sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$x + \frac{b}{2a} = \pm\frac{\sqrt{b^2 - 4ac}}{2a}$$

$$x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$\boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

(i) $x^2 - 9 = 0$

i.e. $x^2 + 0x - 9 = 0$, so $a = 1,\ b = 0,\ c = -9$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-0 \pm \sqrt{(0)^2 - 4(1)(-9)}}{2(1)}$$

$$= \frac{0 \pm \sqrt{0 + 36}}{2} = \frac{\pm\sqrt{36}}{2} = \frac{\pm 6}{2}$$

$$x = \pm 3$$

$$\boxed{\text{S.S} = \{\pm 3\}}$$

(ii) $2x^2 + 5x + 1 = 0$

$a = 2,\ b = 5,\ c = 1$

$$x = \frac{-5 \pm \sqrt{(5)^2 - 4(2)(1)}}{2(2)}$$

$$= \frac{-5 \pm \sqrt{25 - 8}}{4}$$

$$x = \frac{-5 \pm \sqrt{17}}{4}$$

$$\boxed{\text{S.S} = \left\{ \frac{-5 \pm \sqrt{17}}{4} \right\}}$$

(iii) $x^2 - 23x - 24 = 0$

$a = 1,\ b = -23,\ c = -24$

$$x = \frac{-(-23) \pm \sqrt{(-23)^2 - 4(1)(-24)}}{2(1)}$$

$$= \frac{23 \pm \sqrt{529 + 96}}{2} = \frac{23 \pm \sqrt{625}}{2}$$

$$x = \frac{23 \pm 25}{2}$$

$$x = \frac{23 + 25}{2} = \frac{48}{2} = 24 \quad \Big| \quad x = \frac{23 - 25}{2} = \frac{-2}{2} = -1$$

$$\boxed{\text{S.S} = \{24, -1\}}$$

(iv) $(x + 1)^2 = (2x - 1)^2$

$$(x)^2 + (1)^2 + 2(x)(1) = (2x)^2 + (1)^2 - 2(2x)(1)$$

$$x^2 + 1 + 2x = 4x^2 + 1 - 4x$$

$$0 = 4x^2 - x^2 - 4x - 2x + 1 - 1$$

$$0 = 3x^2 - 6x$$

$$3x^2 - 6x = 0$$

$a = 3,\ b = -6,\ c = 0$

$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(3)(0)}}{2(3)}$$

$$= \frac{6 \pm \sqrt{36-0}}{6} = \frac{6 \pm \sqrt{36}}{6}$$

$$x = \frac{6 \pm 6}{6}$$

$$x = \frac{6+6}{6} = \frac{12}{6} = 2 \quad \Big| \quad x = \frac{6-6}{6} = \frac{0}{6} = 0$$

$$\boxed{\text{S.S} = \{2, 0\}}$$

(v) $\frac{x+1}{2} - \frac{x(x+2)}{3} = 0$

Multiply 6 on both sides (L.C.M of 2 and 3 is 6):

$$6 \times \frac{x+1}{2} - 6 \times \frac{x(x+2)}{3} = 0 \times 6$$

$$3(x+1) - 2x(x+2) = 0$$

$$3x + 3 - 2x^2 - 4x = 0$$

$$-2x^2 - x + 3 = 0$$

$$-(2x^2 + x - 3) = 0$$

$$2x^2 + x - 3 = 0$$

$a = 2,\ b = 1,\ c = -3$

$$x = \frac{-1 \pm \sqrt{(1)^2 - 4(2)(-3)}}{2(2)} = \frac{-1 \pm \sqrt{1+24}}{4} = \frac{-1 \pm \sqrt{25}}{4}$$

$$x = \frac{-1 \pm 5}{4}$$

$$x = \frac{-1+5}{4} = \frac{4}{4} = 1 \quad \Big| \quad x = \frac{-1-5}{4} = \frac{-6}{4} = -\frac{3}{2}$$

$$\boxed{\text{S.S} = \left\{1, -\frac{3}{2}\right\}}$$

(vi) $(x-2)(x-6) = (2x+1)(x+1)$

$$x(x-6) - 2(x-6) = 2x(x+1) + 1(x+1)$$

$$x^2 - 6x - 2x + 12 = 2x^2 + 2x + x + 1$$

$$x^2 - 8x + 12 = 2x^2 + 3x + 1$$

$$0 = 2x^2 - x^2 + 3x + 8x + 1 - 12$$

$$0 = x^2 + 11x - 11$$

$$x^2 + 11x - 11 = 0$$

$a = 1,\ b = 11,\ c = -11$

$$x = \frac{-11 \pm \sqrt{(11)^2 - 4(1)(-11)}}{2(1)}$$

$$= \frac{-11 \pm \sqrt{121 + 44}}{2}$$

$$x = \frac{-11 \pm \sqrt{165}}{2}$$

$$\boxed{\text{S.S} = \left\{ \frac{-11 \pm \sqrt{165}}{2} \right\}}$$

Graphing Method — Basic Concept

1. Types of graph:

- **Linear equation graph** (e.g. $x + 2 = 0$): a **straight** line.
- **Quadratic equation graph**: a **curve** called a **parabola**.
  - If a is positive: the parabola opens **upwards** (e.g. $x^2 + 6x + 9 = 0$).
  - If a is negative: the parabola opens **downwards** (e.g. $-x^2 + 6x + 9 = 0$).

2. How to draw graph? (Steps)

1. Write the general form (standard form) of the quadratic equation.
2. Put it equal to y.
3. Put some values of x to get the values of y. To choose the values of x, first find the vertex and take values on both sides of it.

Vertex (turning point): The vertex of a parabola is the point where the parabola changes direction. It is either the lowest point (minimum) of a parabola that opens upwards, or the highest point (maximum) of a parabola that opens downwards.

$$\text{Vertex:} \qquad x = -\frac{b}{2a}, \quad \text{then put this } x \text{ in the equation to get } y.$$

3. How to find roots from graph (parabola)?

The roots are the points where the parabola cuts/crosses the **x-axis**.

- If the parabola cuts/touches the x-axis at **one point**: 2 same roots (e.g. $\{-3, -3\}$).
- If the parabola cuts the x-axis at **two points**: 2 different roots (e.g. $\{-1, 2\}$).

Question 5

Solve $x^2 + 6x = -9$ by graphing and factoring method.

Graphing Method

$$x^2 + 6x = -9$$

$$x^2 + 6x + 9 = 0$$

$$y = x^2 + 6x + 9$$

Vertex: $a = 1,\ b = 6,\ c = 9$

$$x = -\frac{b}{2a} = -\frac{6}{2(1)} = -3$$

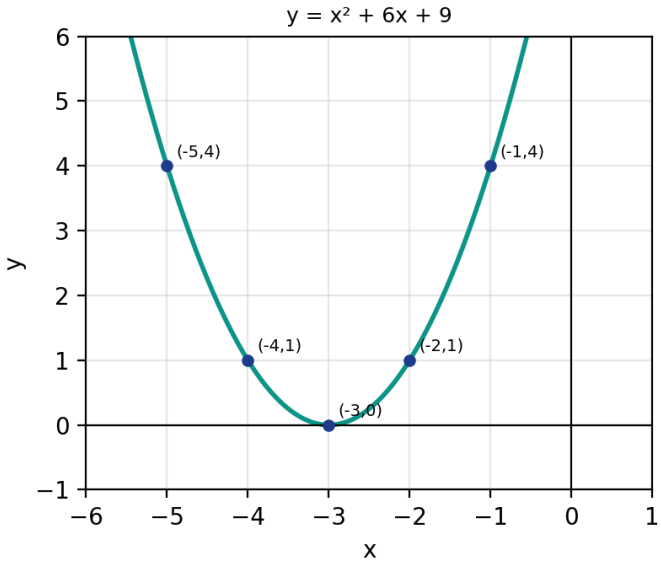
$$y = (-3)^2 + 6(-3) + 9 = 9 - 18 + 9 = 18 - 18 = 0$$

So take values of x around -3: $-5, -4, -3, -2, -1$

$x = -5$	$x = -4$	$x = -2$	$x = -1$
$y = (-5)^2 + 6(-5) + 9 = 25 - 30 + 9 = 4$	$y = (-4)^2 + 6(-4) + 9 = 16 - 24 + 9 = 1$	$y = (-2)^2 + 6(-2) + 9 = 4 - 12 + 9 = 1$	$y = (-1)^2 + 6(-1) + 9 = 1 - 6 + 9 = 4$

x	-5	-4	-3	-2	-1
y	4	1	0	1	4

Plot the points $(-5, 4), (-4, 1), (-3, 0), (-2, 1), (-1, 4)$ and join them with a smooth curve. The parabola opens upwards and touches the x-axis at one point $(-3, 0)$.



Result/Solution: The point where the parabola cuts the x-axis is $x = -3$ (two same roots).

$$\boxed{\text{S.S} = \{-3, -3\}}$$

Factoring Method

$$x^2 + 6x = -9$$

$$x^2 + 6x + 9 = 0$$

Split the middle term: $9 = 9 \times 1 \times$, 3×3 ✓ and $3 + 3 = 6$

$$x^2 + 3x + 3x + 9 = 0$$

$$x(x + 3) + 3(x + 3) = 0$$

$$(x + 3)(x + 3) = 0$$

$$x + 3 = 0 \quad \Big| \quad x + 3 = 0$$

$$x = -3 \quad \Big| \quad x = -3$$

$$\boxed{\text{S.S} = \{-3, -3\}}$$

Both methods give the same solution.

Question 6

Graph the function $y = x^2 + 2x + 4$ and verify the solution by completing square method.

Graph

$$y = x^2 + 2x + 4$$

Vertex: $a = 1,\ b = 2,\ c = 4$

$$x = -\frac{b}{2a} = -\frac{2}{2(1)} = -1$$

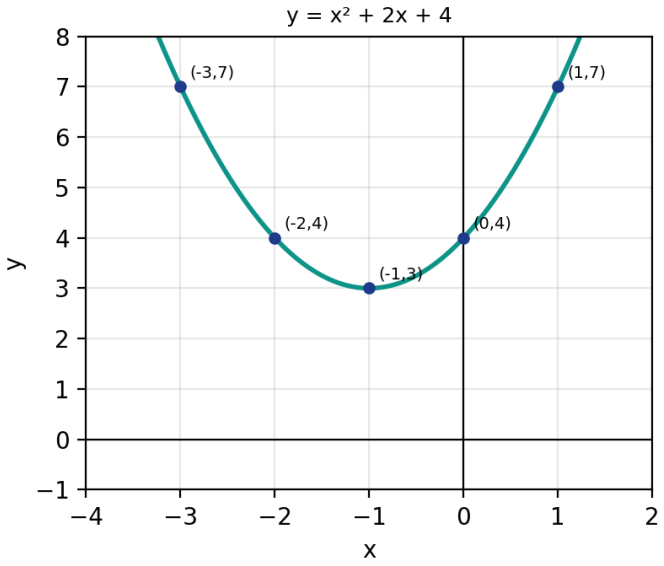
$$y = (-1)^2 + 2(-1) + 4 = 1 - 2 + 4 = 3$$

Take values of x around -1: $-3, -2, -1, 0, 1$

$x = -3$	$x = -2$	$x = 0$	$x = 1$
$y = (-3)^2 + 2(-3) + 4 = 9 - 6 + 4 = 7$	$y = (-2)^2 + 2(-2) + 4 = 4 - 4 + 4 = 4$	$y = (0)^2 + 2(0) + 4 = 4$	$y = (1)^2 + 2(1) + 4 = 1 + 2 + 4 = 7$

x	-3	-2	-1	0	1
y	7	4	3	4	7

Plot the points $(-3, 7), (-2, 4), (-1, 3), (0, 4), (1, 7)$ and join them with a smooth curve. The parabola opens upwards with vertex $(-1, 3)$ and **does not cut the x-axis**, so the equation has no real roots (its roots are complex).



Verifying Solution by Completing Square Method

$$y = x^2 + 2x + 4$$

$$x^2 + 2x + 4 = 0$$

$$x^2 + 2x = -4 \qquad \left(\frac{2}{2} = 1\right)$$

$$x^2 + 2x + (1)^2 = -4 + (1)^2$$

$$(x+1)^2 = -4 + 1$$

$$(x+1)^2 = -3$$

$$\sqrt{(x+1)^2} = \sqrt{-3}$$

$$x + 1 = \pm\sqrt{-3}$$

$$x = -1 \pm \sqrt{-3}$$

$$= -1 \pm \sqrt{3 \times (-1)} = -1 \pm \sqrt{3}\sqrt{-1}$$

$$x = -1 \pm \sqrt{3}\, i \qquad (\because i = \sqrt{-1})$$

$$\boxed{\text{S.S} = \{-1 \pm \sqrt{3}\, i\}}$$

The roots are complex (not real), which verifies that the graph does not cut the x-axis.

Question 7

Explain each term: (i) Solution (ii) Root (iii) Zero of a function (iv) x-intercept

(i) Solution

The values of variable for which an equation becomes a true sentence are called solution.

OR A solution is a value that, when substituted into an equation, makes the equation true.

Example: $2x + 3 = 7$, solution is $x = 2$.

$$2(2) + 3 = 7 \;\Rightarrow\; 4 + 3 = 7 \;\Rightarrow\; 7 = 7 \quad \text{True}$$

Suppose we put $x = 1$: $2(1) + 3 = 7 \Rightarrow 2 + 3 = 7 \Rightarrow 5 \neq 7$, so 1 is not a solution of that equation.

Note: The term "solution" can be used for any type of equation, including those where the result is not zero.

(ii) Root

Roots are specifically the values of the variable that, when substituted into the equation, result in the equation being equal to **zero**.

Example: $3x^2 + 6x - 9 = 0$; roots are $x = 1$ and $x = -3$.

Put $x = 1$: $3(1)^2 + 6(1) - 9 = 3 + 6 - 9 = 9 - 9 = 0$ ✓

(iii) Zero of a Function

- A zero of a function $f(x)$ is a value of x for which $f(x) = 0$.
- In simple words, it is the value of x where the function's graph cuts or touches the x-axis.
- Zeros of a function are also called roots of the function.

Example: $y = f(x) = 3x^2 + 6x - 9$

Put $x = 1$: $f(1) = 3(1)^2 + 6(1) - 9 = 3 + 6 - 9 = 9 - 9 = 0$

So $x = 1$ (and similarly $x = -3$) is a zero of the function; the graph cuts the x-axis at these points.

(iv) x-intercept

An x-intercept is the point where the graph of a function intersects (crosses) the x-axis.

- At this point, the y-coordinate is zero.
- It is represented as $(x, 0)$, e.g. $(1, 0)$ and $(-3, 0)$.
- This x-intercept is also the root of that equation.

Question 8

Can a quadratic equation have more than two solutions? Why or why not?

Answer No. 1

- **No**, a quadratic equation can't have more than two solutions.
- A quadratic equation can have **only two** solutions.

Reason:

- In algebra, the number of solutions depends on the degree (highest power of the variable): number of solutions $\propto$ degree.
- As a quadratic equation $ax^2 + bx + c = 0$ has degree 2, that's why a quadratic equation can have only two solutions.

Answer No. 2

No, a quadratic equation can have only two solutions because of degree 2.

Question 9

Plans for a rectangular ice-skating rink that is 30 m by 60 m have to be revised to double the area of the playground by adding strips of the same width to a side and an end of the area to form a rectangle. Find the width of the strips. What are the dimensions of the playground now?

Before: Length = 60 m, Width = 30 m

$$\text{Area} = 60 \times 30 = 1800 \text{ m}^2$$

To find: Width of strip $= x = ?$

New length $= 60 + x$ New width $= 30 + x$ (new dimensions)

New area (double) $= 2 \times 1800 = 3600 \text{ m}^2$

$$\text{length} \times \text{width} = 3600$$

$$(60 + x)(30 + x) = 3600$$

$$60(30+x)+x(30+x)=3600$$

$$1800+60x+30x+x^2-3600=0$$

$$x^2+90x-1800=0$$

$a=1,\ b=90,\ c=-1800$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{-90\pm\sqrt{(90)^2-4(1)(-1800)}}{2(1)}$$

$$=\frac{-90\pm\sqrt{8100+7200}}{2}=\frac{-90\pm\sqrt{15300}}{2}$$

$$x=\frac{-90\pm123.69}{2}$$

$$x=\frac{-90+123.69}{2}=16.85\text{ m}\quad\Big|\quad x=\frac{-90-123.69}{2}=-106.85\text{ m}$$

Length can't be negative, so $x=16.85$ m.

- Width of strip $=x=16.85$ m
- New length $=60+x=60+16.85=76.85$ m
- New width $=30+x=30+16.85=46.85$ m

Result: The width of the strips added is approximately **16.85 m**. The new dimensions of the playground are approximately **76.85 m by 46.85 m**.

Question 10

If a car has an initial speed of 20 m/s and a constant acceleration of 2 m/s², determine the amount of time it takes to travel 145 m. (Use: $S=v_it+\frac{1}{2}at^2$)

$$v_i=20\text{ m/s},\quad a=2\text{ m/s}^2,\quad S=145\text{ m},\quad t=?$$

$$S=v_it+\frac{1}{2}at^2$$

$$145=20t+\frac{1}{2}\times2t^2$$

$$145=20t+t^2$$

$$0=t^2+20t-145$$

$$t^2+20t-145=0$$

$a=1,\ b=20,\ c=-145$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-20 \pm \sqrt{(20)^2 - 4(1)(-145)}}{2(1)}$$

$$= \frac{-20 \pm \sqrt{400 + 580}}{2} = \frac{-20 \pm \sqrt{980}}{2}$$

$$t = \frac{-20 \pm 31.30}{2}$$

$$t = \frac{-20 + 31.30}{2} = 5.65 \text{ s} \quad \Big| \quad t = \frac{-20 - 31.30}{2} = -25.65 \text{ s}$$

Time can't be negative, so $t = 5.65$ s.

Result: It takes approximately **5.65 seconds** for the car to travel 145 meters.